

Course Outline

Data Structures CS-2001

Semester Fall-2025, Section-BCS-3C, BSE-3A & BSE-3B

Instructor: Razi Uddin
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Office Hours: Monday 2:30 to 3:30
Wednesday 2:30 to 3:30

Credit 3

Prerequisite Object-Oriented Programming

Hours:

Course Objectives:

CS2001 is a core Computer Science course with Object Oriented Programming as its prerequisite. The course learning outcomes are:

- Demonstrate basic concepts of data structure and algorithms.
- Evaluate different data structures in terms of memory complexity and time requirement
- Design appropriate data structures to solve real-world problems related to the program
- Determine bugs in programs and recognize required operations with data structures

Text Book:

Any one of these books is recommended as a textbook:

- Mark Allen Weiss, *Data structures and algorithm analysis*, Pearson Education, 2007.
- Adam Drozdek, *Data structures and algorithms in C++*, Course Technology, 2004.
- Nell Dale, *C++ Plus Data Structures*, 3rd Edition, Jones and Bartlett, 2003.
- Michael T. Goodrich, Roberto Tamassia, and David M. Mount, *Data structures and algorithms*, 2nd Edition, John Wiley & Sons, 2011.

LECTURES	TOPICS
1	Introduction
2	Time Complexity Analysis and Asymptotic Bounds
5	Linked Lists Review of pointers Singly linked lists, doubly linked lists, circular lists, and corresponding iterators
2	Stacks and Queues
MIDTERM 1	
2	Recursion
3	Trees Binary trees and their traversals Binary search trees (Insertion, Deletion, and Search)
3	Height Balanced Binary Search Trees (AVL Trees)
2	Heaps and heap sort
MIDTERM 2	
1	Data compression and Huffman coding
2	Hashing Hash tables and hash functions Collision resolution
5	Graph data structure, Breadth-first search and Depth-first search

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Tentative Grading Scheme:

Assignments **(10 %)**

Quizzes **(10 %)**

Midterms **(30 %)**

Final Exam **(50 %)**

Homework **(0%)**

Important Instructions:

- Quizzes may be announced or surprise
- There will be no makeup quiz
- Minimum requirement to pass this course is to obtain at least 50% marks.
- All assignments and coursework must be done individually. **Plagiarism** in any work (Quiz, Assignment, Midterms, and Final Exam) from any source (ChatGPT/Internet or a Student) will result in an **F** grade.
- No Late assignment submissions will be accepted.
- All the FAST School of Computing's grading policies apply.